5.2.4 Gaussian elimination via Gauss transforms

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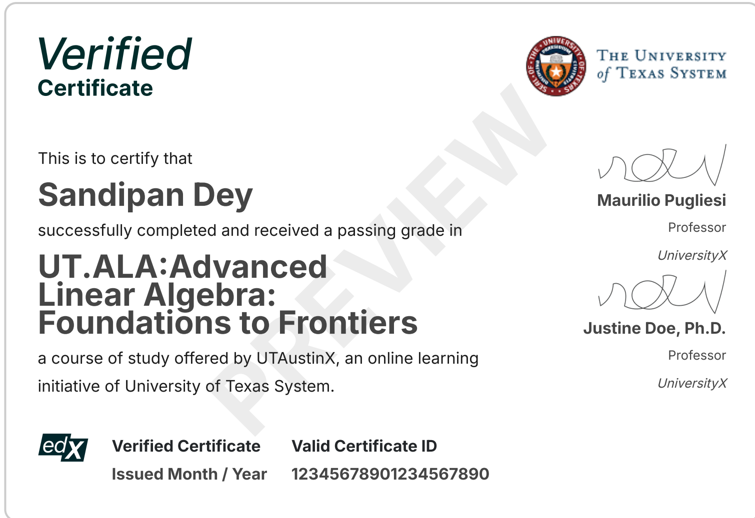

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Now the

purpose of the game is to introduce zeros here. So we would like this result

here to be 0. And it's not hard to see that if you pick l_21 to be a_21

divided by alpha _11 then this expression becomes 0. Where have you seen

that before? That's the computation of the

multipliers. Okay? And then once you have done that, notice that this is the same

as A_22 minus l_21 a 12 transpose. And

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Reading assignment

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Read Unit 5.2.4 of the notes. [LINK]

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Homework 5.2.4.1

1/1 point (graded)

Evaluate

$$\left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array}\right)\left(\begin{array}{c|cc} A_{00} & a_{01}^T & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array}\right)$$

where I_k is the $k \times k$ identity matrix and A_0 has k rows.

- ( ) $\left(\begin{array}{c|cc} A_{00} & a_{01}^T & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} + \alpha_{11} l_{21} & A_{22} + l_{21} a_{12}^T \end{array}\right)$
- ( ) $\left(\begin{array}{c|cc} A_{00} & a_{01}^T & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & 0 & A_{22} - l_{21} a_{12}^T \end{array}\right)$
- (•) $\left(\begin{array}{c|cc} A_{00} & a_{01}^T & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} - \alpha_{11} l_{21} & A_{22} - l_{21} a_{12}^T \end{array}\right)$

✔

If we compute

$$\left(\begin{array}{c|cc} A_{00} & a_{01}^T & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & \widehat{a}_{21} & \widehat{A}_{22} \end{array}\right) := \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array}\right)\left(\begin{array}{c|cc} A_{00} & a_{01}^T & A_{02} \\ \hline 0 & \alpha_{11} & a_{12}^T \\ 0 & a_{21} & A_{22} \end{array}\right),$$

how should l_{21} be chosen if we want $\widehat{a}_{21}$ to be a zero vector?

- ( ) $l_{21} = -a_{21}/\alpha_{11}$
- (•) $l_{21} = a_{21}/\alpha_{11}$
- ( ) $l_{21} = \alpha_{11} a_{21}$

✔

Solution:

For a complete solution, see the notes .

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Homework 5.2.4.2

1/1 point (graded)

Show that

$$\left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & -l_{21} & I \end{array}\right)^{-1} = \left(\begin{array}{c|cc} I_k & 0 & 0 \\ \hline 0 & 1 & 0 \\ 0 & l_{21} & I \end{array}\right)$$

where I_k denotes the $k \times k$ identity matrix.

☑ done

✔

Solution:

For a complete solution, see the notes .

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Homework 5.2.4.3

1/1 point (graded)

Let

$$\tilde{L}_{k-1} = L_0^{-1} L_1^{-1} \ldots L_{k-1}^{-1} = \left(\begin{array}{c|cc} L_{00} & 0 & 0 \\ \hline l_{10}^T & 1 & 0 \\ L_{20} & 0 & I \end{array}\right) \quad \text{and} \quad L_k^{-1} = \left(\begin{array}{c} I_k \\ \hline 0 \\ 0 \end{array}\right.$$

where L_{00} is a $k \times k$ unit lower triangular matrix. Show that

$$\tilde{L}_k = \tilde{L}_{k-1} L_k^{-1} = \left(\begin{array}{c|cc} L_{00} & 0 & 0 \\ \hline l_{10}^T & 1 & 0 \\ L_{20} & l_{21} & I \end{array}\right).$$

☑ done

✔

Solution:

For a complete solution, see the notes .

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